4

Disease control and lesion prevention

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4.1 Introduction

From the previous chapter it can be seen that in all cases of caries and tooth wear, minimum intervention disease control and lesion prevention are key aspects of the management strategy, often commencing in the stabilization phase of the care plan, but continuing throughout the full course of treatment and beyond in order to maintain lifelong oral health.

4.1.1 Disease control

Neither the dentist, as part of the oral healthcare team, nor the patient has the power to *prevent* the caries or tooth-wear process. These ubiquitous processes occur at the ionic, metabolic, and microscopic level at the tooth surface/biofilm interface, and are made pathological by other factors in combination. If these factors are controlled or modified by the patient (with help from the oral healthcare team), then the processes can be regulated. The term *primary prevention* is sometimes used in this context.

4.1.2 Lesion prevention

The term *prevention* has been commonly used, but actually it is only the manifestation of the pathological process (i.e. the lesion in caries or tooth wear) that can be prevented if the disease process is controlled. The term *secondary prevention* has been used in this context to slow down or stop (arrest) incipient, progressing lesions. *Tertiary prevention* is a term sometimes used to describe the care offered to the patient in an attempt to control or reduce the pattern of future disease.

4.2 Caries control (and lesion prevention)

4.2.1 Categorizing caries activity and risk status

On the basis of the history and examination, the patient may be allocated to one of the following in terms of caries activity/risk status:

- ***Caries inactive/caries controlled/low risk:*** no active lesions and no history of recurrent active restorations in the past 2 to 3 years. A level of control (maintenance) is still required to remain in this stable condition.
- ***Caries active/modifiable risk factors/moderate risk*** (plaque control, fluoride, diet): presence of active lesions and a yearly increment of more than two new/progressing/filled lesions in the preceding 2 to 3 years. Caries control may be achieved by changing/modifying risk factors.
- ***Caries active/unmodifiable or unidentifiable risk factors/high risk*** (dry mouth, medications): this category will always be high risk, although it may still be possible to control caries by optimal moderation of such risk factors. Presence of active lesions and a yearly increment of more than two new/progressing/filled lesions in the preceding 2 years.

The aim is to help the patient to change their risk factors so that, at recall, the caries activity status may be deemed to have changed because there are no new active lesions and/or lesions previously judged to be active are now deemed to be arrested. The correlation between current risk status of the patient and the presence of active lesions (and their stage of progress—a *caries risk matrix*) can lead to an assessment being made by the oral healthcare team, in conjunction with the patient, of their likelihood of suffering from new disease in the future. Table 4.1 shows a flowchart of potential care plans depending on active lesion presence and the patient's caries risk. Note that in all cases disease control is paramount in preventing further lesion development, but there are different levels of control that can be offered depending on the activity/risk status—*standard* and *active* care (see Table 4.2).

4.2.2 Standard care (non-operative, preventive therapy): low-risk, caries-controlled, disease-inactive patient

The standard care regimen is carried out wholly by the patient on the advice of the suitably trained oral healthcare team dental care professional (dentist, oral health educator, hygienist, or therapist).

Plaque control

Caries lesions form as a result of the metabolic events in the dental plaque biofilm. Thus plaque control is the logical cornerstone of non-operative therapy. Teeth should be brushed with a fluoride-containing toothpaste, as the latter interferes with growth and ecology of the biofilm, and fluoride application retards lesion progression.

The preventive action of tooth brushing can be maximized if the following principles are followed:

- Brushing should start as soon as the first deciduous tooth erupts.
- Brush twice daily, last thing at night, and at one other time each day.
- Discourage rinsing with large amounts of water after brushing—'spit, don't rinse' is the correct advice.
- Evidence exists that using electric rotating/oscillating toothbrushes can be more effective in plaque biofilm removal, but this will be patient dependent.
- Children under 3 years of age should use a toothpaste containing no less than 1000 ppm (parts per million) of fluoride.
- Children between 3 and 6 years of age are likely to swallow toothpaste, and this may cause fluorosis. To prevent this they should use only a smear of paste (a pea-sized amount) on the brush head, and should not be allowed to eat or lick toothpaste from the tube.
- Currently it is recommended that over the age of 3 years, family fluoride toothpaste (1350–1500 ppm fluoride) should be used.

Table 4.1 A care plan flowchart for caries management showing the interaction of caries susceptibility, presence/absence of lesions, and an emphasis on minimally invasive dentistry for the operative treatment of individual lesions with suitable recall intervals

Identify (Chapter 2) Caries susceptibility	**Active lesions**			**No/inactive (arrested) lesions**	
	Cavitated (irreversible)	Non-cavitated (reversible)			
	High/low caries risk 3, 4 (mICDAS)	High risk 0, 1, 2	Low risk 0, 1, 2	High risk	Low risk
Control (prevent) (Chapter 4) Preventive care and patient motivation	**SC + active care plus:** Pit and fissure sealants Patient motivation	**SC + active care:** Remin Fluoride CPP-ACP PMTC Patient motivation	**SC + active care:** Remin Fluoride CPP-ACP PMTC	**SC + active care:** Fluoride CPP-ACP PMTC Patient motivation	**SC + maintenance** Oral hygiene Diet control Fluoride toothpaste
Restore (Chapters 5 and 8)	Transitional restorations: GIC Long-term direct restorations: GIC/composite/amalgam	Preventive resin restorations	–	–	–
Recall (Chapter 9)	2–6 months	3–6 months	6 months	3–6 months	12–18 months

The column on the left indicates where in this book those topics are discussed in more detail. See Table 4.2 for explanations of standard care and active care regimens.

CPP-ACP, casein phosphopeptide–amorphous calcium phosphate; PMTC, professional mechanical tooth cleaning; SC, standard care.

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Table 4.2 Features of preventive care regimens, standard and active care, to control caries and prevent lesions from developing or progressing

Standard care (patient-led, non-operative) **Inactive/controlled, low-risk patient**	**Active care (dentist-led, operative/non-operative)** **Active/uncontrolled, high-risk patient**
Plaque control (toothbrush and toothpaste, flossing) Use of **fluoride** (toothpaste, water) **Dietary** modification **Continued patient motivation** paramount to prevent onset of disease	**STANDARD CARE +:** **Decontamination** procedures (PMTC, transitional restorations, chlorhexidine) **Remineralization** procedures: • Fluoride (high-concentration toothpaste, mouthwashes, topical varnishes) • Remineralizing pastes/solutions (e.g. CPP-ACP, bioactive glasses) **Managing hyposalivation** (medications, saliva substitutes) **Fissure sealant restorations**

- Children need to be helped and supervised by an adult when brushing.
- The occlusal surface of erupting molars should be individually brushed with the brush placed at right angles to the arch.
- Dependent adults should be helped with tooth cleaning.

Oral hygiene instruction (OHI) should be general to the whole mouth, and site-specific to the particular lesion. The patient should be aware of the problem areas, seeing these in their mouth (via mirrors/images from intra-oral cameras) and/or on a radiograph (with an explanation at a suitable level provided). The following may be helpful with regard to tooth brushing:

- The patient should bring their brush and toothpaste to each appointment they attend.
- Check the toothpaste for fluoride content and the brush head quality to ensure it is not worn.
- Disclose the mouth with a suitable plaque-staining dye so that this can be clearly seen by the patient.
- Assess whether the patient (or parent/carer) can remove the plaque or whether the technique/brush should be altered. For example, arthritic patients may not be able to manipulate the normal handle of a toothbrush, so silicone impression material or cold-cure acrylic may be used to thicken the grip to aid manual dexterity. Rotating/oscillating electric toothbrushes may also be recommended.

- Is thorough brushing in the surgery causing gingival bleeding? If so, does the patient realize that this indicates gingivitis caused by dental plaque?
- If active lesions are present, is the patient aware of where they are and able to remove disclosed plaque from them?

Where active proximal lesions are present, either in the enamel or on the root surface, an interdental cleaning aid will be needed. In young patients, lesions are best cleaned with floss, whereas interdental brushes are preferred for cleaning larger interdental spaces following gingival recession. The following may be helpful with regard to interdental cleaning:

- Advice given must be site-specific.
- Examining the tape or brush after use may show the plaque that has been removed, and this can be a useful motivating factor.
- A special holder for the floss or brush may help the patient.
- Super-floss may be used to clean around bridge pontics or crown margins.
- If the gingivae bleed, the relevance of this should be explained to the patient. If bleeding persists for days after effective cleaning is instituted, this may indicate a cavity or plaque-retentive feature is present that is preventing the patient from removing the plaque. A restoration may be needed to restore tooth integrity to allow the surface to be cleaned.

Use of fluoride

Fluoride delays lesion progression by being incorporated physico-chemically into the hydroxyapatite lattice structure and by inhibiting carbohydrate metabolism within the plaque bacteria (enolase and phosphoenolpyruvate (PEP)-phosphotransferase system inhibition). Vehicles for fluoride include the following:

- *Water*: to date, approximately 15% of the UK population have access to 1 ppm F– water
- *Toothpaste*: fluoride toothpaste is cheap, requires minimal patient cooperation, and enhances patients' appreciation of their essential role in caries control. See page 54 for advice on fluoride concentration and toothpaste use.

The choice of vehicle is not crucial, but it must be combined with improvement in oral hygiene. It is important that the patient accepts the mode of treatment and complies with advice.

Dietary modification

The evidence that the frequency and amount of sugar consumption are linked to caries is irrefutable. Thus an emphasis on diet in caries control would seem logical. Unfortunately, the evidence that it is possible to modify people's diets is lacking! Since the advent of fluoride, the emphasis in caries control has shifted from diet to oral hygiene in conjunction with a fluoride-containing toothpaste. However, this does not preclude the dental care professional from giving dietary advice to all patients. Reducing the amount and frequency of sugary food intake can reduce dental caries and could help weight control. All health professionals have a responsibility to give advice on diet, in the same way as they have a responsibility to give advice on smoking cessation.

No change in diet should be advised for the caries-inactive/controlled patient, but the dentist/team should make the patient aware of how an adverse change in diet and/or salivary flow could pose a future problem, especially if oral hygiene is poor. All patients should be aware of the link between saliva, sugar, and caries.

All of the aspects of *standard care* for controlling disease and preventing lesions require patient cooperation and motivation. The dental team is the adviser, and the patient is the executor of their own preventive care plan.

4.2.3 Active care: ! high risk/uncontrolled, disease-active patient

This regimen includes all aspects of standard care *plus* those listed in Table 4.2, where the dental team may have to take on some of the treatment as well as an advisory role.

Decontamination procedures

Professional mechanical tooth cleaning (PMTC)

This is carried out by members of the oral healthcare team (dentist, hygienist, therapist) using hand scaler/ultrasonic/air-polishing instrumentation to remove gross calculus deposits, and rotating brushes with prophy paste to debride mechanically particularly thick, tenacious plaque deposits on the tooth surfaces. This will then allow the patient to clean more effectively with their toothbrush while enabling the clinician to examine the tooth surfaces directly.

Transitional (stabilizing) restorations

In some cases (e.g. rampant caries), early stabilizing minimally invasive adhesive restorations (high-viscosity glass ionomer cements (GICs)/resin modified (RM)-GICs; see Chapters 7 and 8) can be placed, not only to strengthen remaining tooth structure but also to remove the carious biomass, which acts as a bacterial reservoir and is impossible to clean effectively with conventional oral hygiene procedures (see Figure 4.1).



Figure 4.1 An anterior view of a 24-year-old caries-active/uncontrolled, high-risk patient with rampant caries. The large cavitated lesions are covered with a tenacious plaque biofilm. In such patients, transitional (stabilizing) restorations might be necessary to remove the bacterial reservoir and to recontour the smooth tooth/restoration surfaces to facilitate improved oral hygiene procedures.

Q4.1: Look at Figure 4.1. What might be the possible aetiology of this degree of caries attack?

Antimicrobial agents

The present understanding is that dental caries is the result of an imbalanced enrichment of plaque, driven by nutrients that encourage acidogenic and acid-tolerant bacteria (the ecological plaque hypothesis). This may justify the inclusion of some antibacterial agents in a caries prevention regime. These could be incorporated in a toothpaste or mouth rinse. Whatever the delivery method, they should be aimed at:

- minimizing plaque formation
- reducing the growth of specific bacteria
- modifying enzymes that control the production of acid.

Chlorhexidine (CHX) rinses

There is little clinical evidence for the use of chlorhexidine in the medium- to long-term control of caries. CHX (> 0.12%) can affect plaque biofilm adherence and bacterial diversity on a short-term basis, and can be used following oral surgical procedures when normal oral hygiene procedures would be difficult to perform due to pain, risk of post-operative bleeding, and limited access for oral hygiene aids. Due to a high substantivity, its efficacy is related to both concentration and frequency of application. Long-term use may lead to altered taste perception, extrinsic dental staining, and possible mucosal irritation, and is therefore not recommended.

Xylitol

Several studies have shown a significant decrease in *Streptococcus mutans* load after 5 weeks of chewing gum containing xylitol as a sweetener. The minimal effective dose was around 6.5 g per day, divided into at least three chewing periods. Xylitol is safe at the previously mentioned recommended dose, its main side effect—along with other sugar alcohols—being a mild laxative effect.

Remineralization procedures

Fluoride

- High-concentration prescription toothpastes can be beneficial for controlling caries (2800/5000 ppm fluoride) in high-risk patients (multiple lesions, dry mouth). This is because it tips the demineralization/remineralization balance in favour of remineralization, and has the following antimicrobial effects:
  - It impairs glycolysis and metabolic processes in plaque.
  - It inactivates metabolic enzymes.
  - It impairs bacterial membrane permeability to ionic transfer.
  - It inhibits the synthesis of extracellular polymers

- Fluoride mouthwash can be prescribed for patients aged ≥ 8 years, for daily (0.05% NaF)/weekly (0.2% NaF) use. Below 8 years of age, these mouthwashes are not advised because there is a risk of sufficient mouthwash to cause fluorosis in the developing dentition. The rinse should be in addition to twice-daily brushing with toothpaste containing at least 1350 ppm fluoride. Rinses require patient compliance and should be used at a different time to tooth brushing in order to maximize the topical effect, which relates to frequency of availability. The product should be rinsed around the mouth for a timed minute. Be aware that some products are astringent and will be uncomfortable for children and painful for those with a dry mouth and thin, friable intra-oral mucosa. The use of a bland, non-alcoholic mouthwash should be advised. The indications for the use of fluoride mouthwash are:
  - patients over 8 years with high caries activity
  - patients with orthodontic appliances, which inevitably encourage plaque accumulation and predispose to carious lesions
  - patients with a dry mouth (xerostomia)
  - patients with developing root caries. In these patients a weekly concentrated mouthwash may be advised for daily use.

- Professionally applied fluoride varnishes/fluids (e.g. Duraphat, Elmex Fluid) with concentrations of 22 600 ppm F or 2.2% F. Systematic reviews of research have shown that fluoride varnish application by dental care professionals can reduce caries increment in the deciduous dentition by a third, and that in the permanent dentition by nearly half. These are impressive reductions in caries, but the dental professional should be aware of the following:
  - Professional application must be repeated at 3-monthly (high-risk child/adult) to 6-monthly (3 years to teens, adults with dry mouth/active caries) intervals to be effective, and this is inevitably costly.
  - The emphasis switches to the care being provided by the professional rather than by the patient. This is important, as it is essential that the patient is motivated and fully appreciates their own responsibility for maintaining their own oral health. Fluoride is not a panacea for controlling dental disease.
  - The concentration of fluoride is high, and this means that the varnish is potentially toxic to a small child if swallowed. The maximum dose advised for use in the primary dentition is 0.25 ml, and in the mixed dentition is 0.5 ml.
  - The varnish should be applied to isolated, clean, dry teeth. An ideal time to apply varnish is therefore when the teeth are being examined for carious lesions, because in order to detect lesions, the teeth must be isolated, clean, and dry (see Chapter 2). Once the charting is complete, it takes only seconds to apply varnish to fissures, over contact points, cervical margins buccally and lingually, and exposed root surfaces.

Topical remineralizing agents

Much research is being carried out into the development of calcium- and phosphate-rich ionic solutions incorporated, with or without fluoride, into toothpastes, mouth rinses, varnishes, chewing gum, and topical creams, that can act potentially as a reservoir for these mineralizing ions, thereby encouraging surface new mineral deposition. These have the potential to help to combat and prevent caries and dental erosion. An elevated concentration of calcium in plaque biofilm fluid also facilitates a higher build-up of fluoride in this reservoir. In addition, it has been demonstrated that there is a symbiotic relationship between calcium and fluoride.CPP-ACP (casein phosphopeptide–amorphous calcium phosphate) ($Ca_9(PO_4)_6 + H_2O$) is one such stabilized calcium phosphate system, sold under the trade name Recaldent. ACP has a

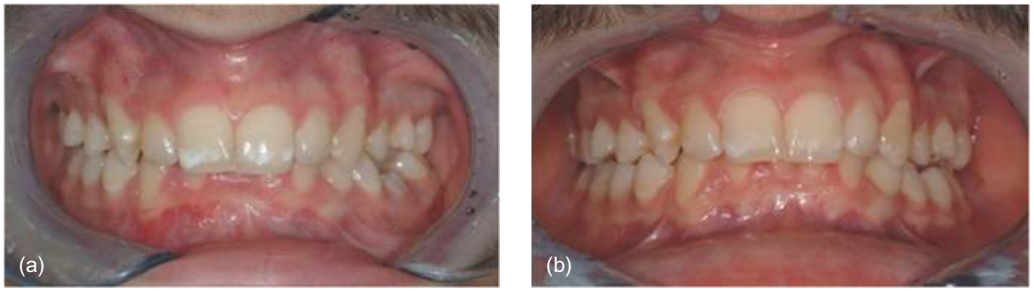


Figure 4.2 **(a)** Patient with chalky, matt fluorotic lesions on the labial surfaces of the upper central incisors (see Chapter 2, Table 2.7). **(b)** The same patient 1 month later, after professional application of topical fluoride/CPP-ACP paste. The lesions are not as clinically evident. (Courtesy of Dr M. Basso.)

Q4.2: How can you distinguish the lesions in Figure 4.2 from white spot carious lesions?

natural predilection to convert to hydroxyapatite (HAP) in wet, ionically favourable environments, and the phosphopeptide chains are thought to help to bind these complexes to the tooth/biofilm surfaces and envelop them, thereby regulating and directing the ultimate precipitation of the HAP where it is needed (i.e. demineralized surfaces). Combinations of CCP-ACP with fluoride have also been developed as pastes and varnishes, and clinical evidence is being gathered as to their overall efficacy in the preventive management of early carious and fluorotic lesions, especially in adults, where the mix of fluoride, calcium, and phosphate ions has a synergistic effect (see Figure 4.2). Current research evidence tends towards a favourable short-term remineralizing effect, but further long-term trials are needed to provide conclusive clinical evidence.

Other calcium- and phosphate-rich formulations include calcium phosphosilicate glass-based dentifrices, functionalized β-tricalcium phosphate, unstabilized amorphous calcium phosphates, nano-hydroxyapatite, and calcium sodium phosphosilicate bioactive glass (NovaMin®), which has been combined with sodium fluoride in a toothpaste. Bioactive glass particles produce crystalline hydroxycarbonate apatite in biological aqueous environments over time, encouraging nucleation sites for further remineralization. Recent *in-vitro* studies have shown how these particles may also be used to surface pre-condition white spot lesions using air abrasion, thus aiding new mineral deposition. However, real-life clinical evidence is lacking for many of these formulations, and more medium- and long-term clinical trial data are required.

Managing hyposalivation

As altered saliva quality and production affect up to 20% of the current UK population, and saliva has an important protective role in the oral cavity, the detection of dry mouth (xerostomia) signs and symptoms in general dental practice is of paramount importance. It should be based on the clinical findings from a targeted examination of the quality and quantity of saliva present and the use of a suitably straightforward clinical assessment system, such as the Challacombe Clinical Oral Dryness Scale (see Chapter 2, Table 2.6, and QR code image 2.2).Controlling caries when the mouth is dry is very difficult. A dry mouth is miserable for the patient, with significant reductions in quality of life indices reported extensively in the literature, and is a concern for the dentist and their oral healthcare team. The long-term management approach for caries control in these high caries risk patients should be as follows:

- Maintain immaculate oral hygiene.
- Prescribe a high fluoride concentration toothpaste (5000 ppm F).
- Prescribe a (non-alcohol-containing) fluoride mouthwash.
- Apply fluoride varnish to existing lesions every 3 months (see previous list).
- Ask the patient to keep a diet sheet and to try to minimize sugar intake.
- Be aware that the patient needs to moisten their mouth frequently, and that plain water is safe.
- Recall the patient every 3 months.
- Saliva production may be stimulated by chewing a xylitol-containing chewing gum, provided there is sufficient salivary gland activity, or pharmacologically.
- In cases of Sjögren's syndrome or post radiotherapy of the head and neck, a saliva substitute may be required, including proprietary dry mouth gels.
- Check that any prescribed antifungal agent does not contain sugar.
- Check with the medical practitioner regarding alternatives to medications that cause hyposalivation (this may be limited, as the medical benefits of the medication will usually outweigh the negative effect of the reduced saliva output).

'Preventive' fissure sealants (FS)

This primary preventive procedure has been included in this section on caries control, as it helps to modify local surface factors that affect the onset and progress of caries and its lesions, namely eradicating caries-susceptible, deep fissures/pits on posterior occlusal surfaces of high-risk patients. There are two types of FS materials:

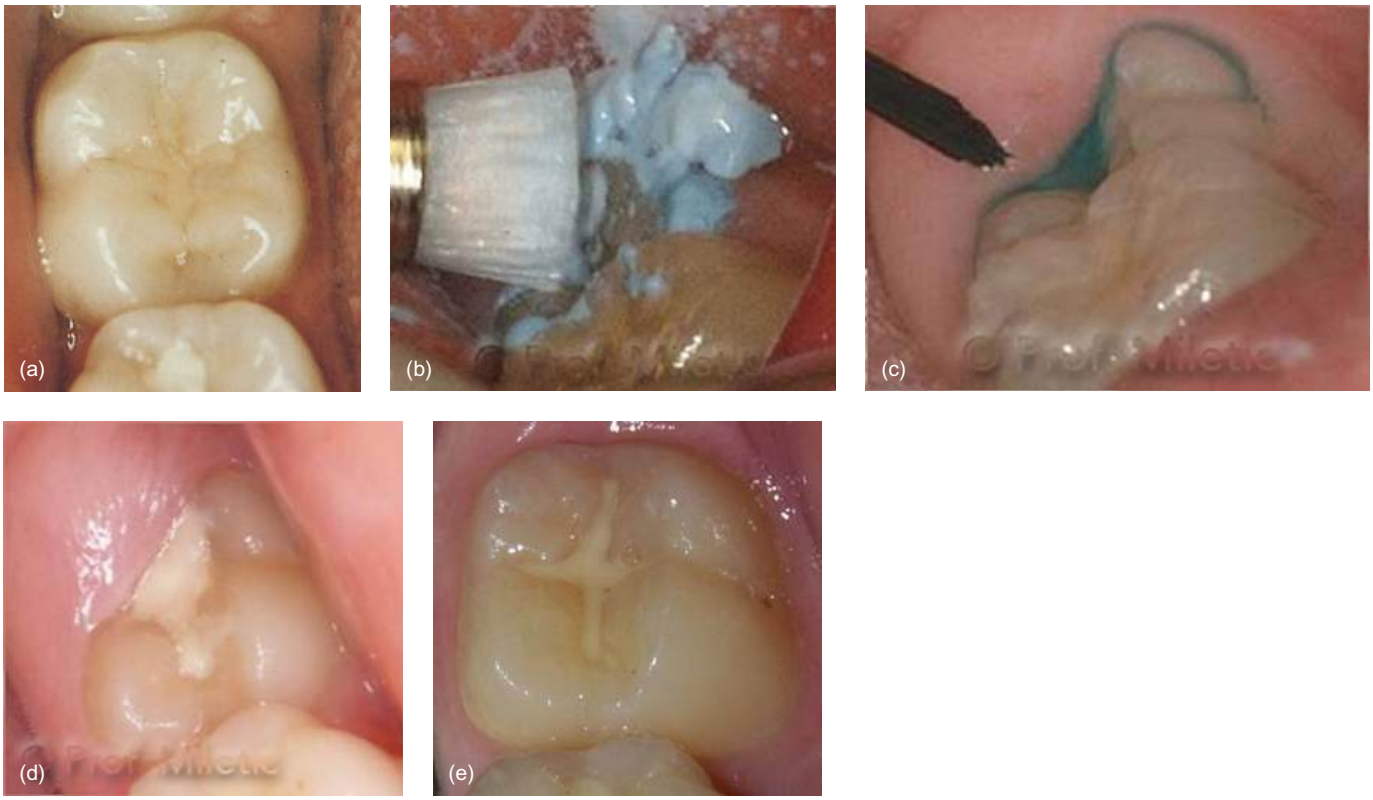


Figure 4.3 **(a)** A glass ionomer fissure sealant on the occlusal surface of a mandibular molar. **(b)** Partially erupted mandibular left second molar in a high caries risk 12-year-old patient with difficult moisture control and access. Occlusal surface is debrided with prophy paste on a rotating bristle brush. **(c)** Ten per cent polyacrylic acid conditioner placed on occlusal surface to remove smear layer. Moisture control is difficult due to the presence of the distal gingival operculum, but was achieved with cotton wool roll isolation (see Chapter 5, Section 5.6). **(d)** The GIC fissure sealant placed (GC Fuji Triage) and **(e)** refurbished 2 years post-eruption in the still caries-free molar. (Figure 4.3 (a) courtesy of the late Dr J. W. McLean; (b)–(e) courtesy of I. Miletic.)

Q4.3:

i. How long would you expect GIC fissure sealants to last? In the patient in Figure 4.3, do you think the FS on the LL7 will need to be replaced indefinitely?

ii. What is the name given to the overlying segment of gingivae on the distal aspect of the occlusal surface on the mandibular molar in Figure 4.3 (d)?

- *Resin-based*: visible light-cured systems based on methacrylate resin composite chemistry (described in Chapter 7). These resins are only lightly filled in order to permit a readily flowable consistency, so creating a void-free infill of deeper fissures. Teeth need to be isolated to obtain effective moisture control, debrided thoroughly, and acid-etched so that the FS is retained micro-mechanically to the enamel surface and light-cured (see Chapter 8, Table 8.2). Care must be taken to ensure the FS is not in direct occlusion, as it will wear or chip readily, with the subsequent risk of increasing plaque accumulation in an uncleansable site.
- *GIC-based:* GICs can be used to bond chemically to the 10% polyacrylic acid-conditioned enamel surface. These materials are more soluble than the resin-based systems, but do leach fluoride ions, which may have a cariostatic effect (see Figure 4.3).

Evidence-based systematic reviews have concluded that there is no clinically significant advantage of one sealant type over another, although it appears that the resin-based sealants show an increased longevity in clinical use when careful application steps are adhered to, and better overall wear resistance. However, in clinical situations where adequate moisture control is difficult to achieve, GIC-based fissure sealants may still be recommended. Both types of FS require careful review and inspection to ensure their long-term integrity. When FS chip or wear away they will need refurbishing, repair, or replacement (see Chapter 9).



Figure 4.4 Mandibular right first and second molars with aged resin-based fissure sealants. (Courtesy of L. Mackenzie.)

Q4.4: Look at Figure 4.4. What concerns you about the appearance of the aged sealants in the LR6 and LR7?

The detailed clinical procedure of placing a preventive fissure sealant is described in Chapter 8 (Table 8.3). The indications are as follows:

- In high-risk, caries-active adolescents, fissure sealing all erupting molars may be advisable as a primary preventive measure.
- In high-risk young adults where there is evidence of caries on molar teeth which need restoration, other susceptible occlusal fissures may be sealed as a precaution.
- In low-risk, caries-inactive patients with good oral hygiene and minimal/no plaque deposits (checked with disclosing solutions), FS are not generally indicated.
- In young patients with deep fissure patterns, who may otherwise be susceptible due to dietary or other local factors changing (e.g. mentally/physically disabled, patient undergoing extensive orthodontic treatment, medically compromised).
- Cleaned teeth which can be easily isolated using a rubber dam (see Chapters 5 and 8). Saliva contamination during placement may lead to premature complete loss of the resin-based fissure sealant or leakage at the margins of a partially debonded restoration, which might then lead to active caries. In such clinical scenarios, GIC-based FS may be advised, but with suitable monitoring.

'Therapeutic' fissure sealants

These have been included here as secondary preventive surface treatments for teeth that have undergone early physical damage due to the caries process. Again, most often placed in the pits and fissures of posterior teeth, these sealants are placed with isolation by rubber dam in ideal conditions, after the incipient lesions present have been debrided thoroughly, perhaps remineralized, but not excavated physically. The sealing action of this minimally invasive tooth surface modification, along with suitable maintenance by the patient, will cause the early lesion to become inactive/arrest. There is some ambiguity in this terminology, as the term 'therapeutic fissure sealant' has also been used by some to describe what are known as *preventive resin restorations*, discussed in the following section.

Preventive resin restoration (PRR)/sealant restoration/invasive/therapeutic fissure sealant

This is placed after the minimally invasive excavation of occlusal caries, up to or just beyond the enamel–dentine junction. In essence, the operator will have undertaken an enamel–dentine 'biopsy' to determine the extent of caries penetration into the tooth. The PRR restores the cavity with a suitable adhesive restorative material (resin composite or GIC) followed by an overlying preventive resin-based fissure sealant which extends on to the unrestored fissures on the remaining debrided and acid-etched sound portion of the occlusal surface, placed as primary prevention for the remaining tooth surface (see Chapter 8, Table 8.3).

4.3 Tooth-wear control (and lesion prevention)

4.3.1 Process

The control and prevention of tooth wear and its lesions are linked intimately with management of the aetiological factors of erosion, attrition, and abrasion discussed in Chapters 1 and 2. The patient's understanding and motivation are vital to the ultimate successful control of tooth wear. The aetiological factors must be identified (by a targeted history and examination) and behaviour modified subsequently (with positive suggestion) in order to remove the causative factor(s). For example, high intake of carbonated drinks/acidic fruit juices over a prolonged period can be adjusted by suggesting that, initially, one in three drinks could be tap water and then, after 1 month, alternate drinks, etc., thereby gradually reducing the primary causative factor.

4.3.2 Lesions

Once structural tooth surface loss has occurred, it cannot be reversed. Early asymptomatic lesions in enamel (or just into dentine) may be left untreated, especially if the causes have been eradicated. The reviewing and recall of patients with pathological tooth wear not requiring immediate operative intervention is discussed in Chapter 9. More extensive tooth loss (accompanied by symptoms including sensitivity, aesthetic considerations, and functional difficulties) can be repaired, depending on the severity, with minimally invasive direct adhesive aesthetic restorations, or with indirect extracoronal restorations (see Figure 4.5). Important factors, including the longevity of the protective restorations placed, and changes in the vertical dimension caused by the tooth surface loss, have to be taken into account in the restorative care plan, details of which are beyond the scope of this book.



Figure 4.5 **(a)** Preoperative anterior view of a patient with erosion–attrition tooth wear caused by gastro-oesophageal reflux disease. **(b)** Anterior view after treatment with metal-ceramic indirect crowns on UR3 to UL3.

Q4.5: Look at Figure 4.5 and answer the following questions.

i. What might the presenting complaints have been?

ii. The long-term tooth loss has caused a loss in occlusal vertical dimension (see Chapter 5). What clues are present in these images to indicate that this process has occurred?

iii. Do you know what might have caused the scar on the anterior mandibular mucosae?

4.4 Suggested further reading and PubMed keywords

www.ncbi.nlm.nih.gov/pubmed/advanced

QR code image 4.1 Try searching the following keywords on PubMed for relevant further reading. You can access PubMed by scanning the QR code image or at the address .

Keywords

dental remineralizing solutions; CPP-ACP.

4.5 Answers to self-test questions

Q4.1: Look at Figure 4.1. What might be the possible aetiology of this degree of caries attack?

A: Long-term poor diet and oral hygiene. This appearance is also a presentation of recreational drug-induced caries in young adults—possibly use of methamphetamine or methadone (a viscous, sugary syrup).

Q4.2: How can you distinguish the lesions in Figure 4.2 from white spot carious lesions?

A: The site—discrete white spot lesions are usually found closer to the enamel–dentine junction and sites associated with plaque stagnation. These lesions are present across the incisal third of the labial surface and not associated with plaque.

Q4.3:

i. How long would you expect GIC fissure sealants to last? In the patient in Figure 4.3, do you think the FS on the LL7 will need to be replaced indefinitely?

A: Depending on patient factors, realistically they would probably last for about 2 years or so. This young patient was initially deemed at high caries risk, so the FS was justified. However, over time the preventive oral care undertaken by the patient will hopefully change their risk status. Indeed, this patient maintained a low-risk status and underwent orthodontic treatment. Therefore the GIC FS was not replaced when it finally failed.

ii. What is the name given to the overlying segment of gingivae on the distal aspect of the occlusal surface on the mandibular molar in Figure 4.3 (d)?

A: Gingival operculum.

Q4.4: Look at Figure 4.4. What concerns you about the appearance of the aged sealants in the LR6 and LR7?

A: In the LR6 the FS appears intact, but there is a grey shadow evident. In a high-risk patient one might be concerned whether this was active caries or not. Other signs, symptoms, investigations, and information gleaned from previous notes would be taken into consideration when deciding whether to monitor or to operatively intervene. In the LR7 the FS has chipped at either end of the horizontal fissure. This would be a concern as it would represent a plaque trap that would be difficult for the patient to clean, and it would need to be at least repaired (see Chapter 9).

Q4.5: Look at Figure 4.5 and answer the following questions.

i. What might the presenting complaints have been?

A: Poor appearance, sharp edges of the incisors cutting the tongue, teeth fracturing, sensitivity.

ii. The long-term tooth loss has caused a loss in occlusal vertical dimension (see Chapter 5). What clues are present in these images to indicate that this process has occurred?

A: Note the step up in the line of the attached gingivae of the mandible subjacent to the lower incisors. This is the classic appearance of *dentoalveolar compensation* that has occurred in an attempt to maintain the vertical relationship between the maxillary and mandibular teeth.

iii. Do you know what might have caused the scar on the anterior mandibular mucosae?

A: Post-operative scar after fixation of fractured mandible with a bone plate several years ago.